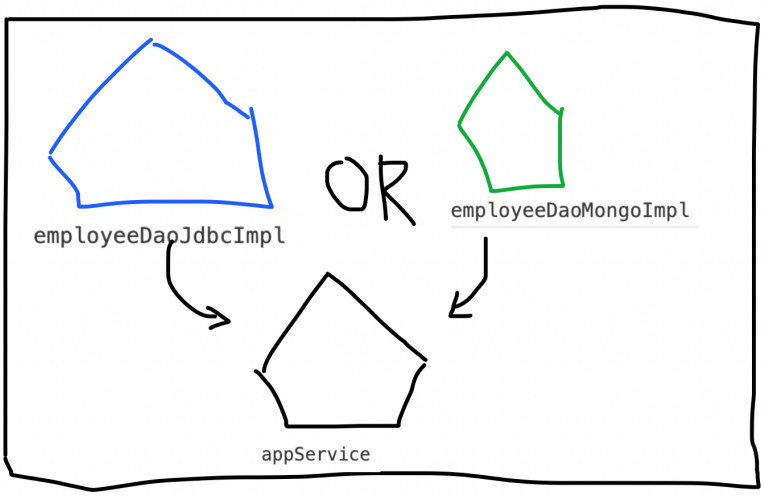
Problem:

```
Field employeeDao in com.cisco.demo.service.AppService required a single bean, but 2 were found:
    - employeeDaoJdbcImpl
    - employeeDaoMongoImpl
```

```java
@Service
public class AppService {
    @Autowired
    private EmployeeDao employeeDao;
```




```
Field employeeDao in com.cisco.demo.service.AppService required a single bean, but 2 were found:
    - employeeDaoJdbcImpl
    - employeeDaoMongoImpl
```

Expand the window

```java
@Entity
@Table(name="customers")
public class Customer {
    @Id
    private String email;

    @Column(name="FNAM", length=100)
    private String firstName;
}
```

customers

email	fnam

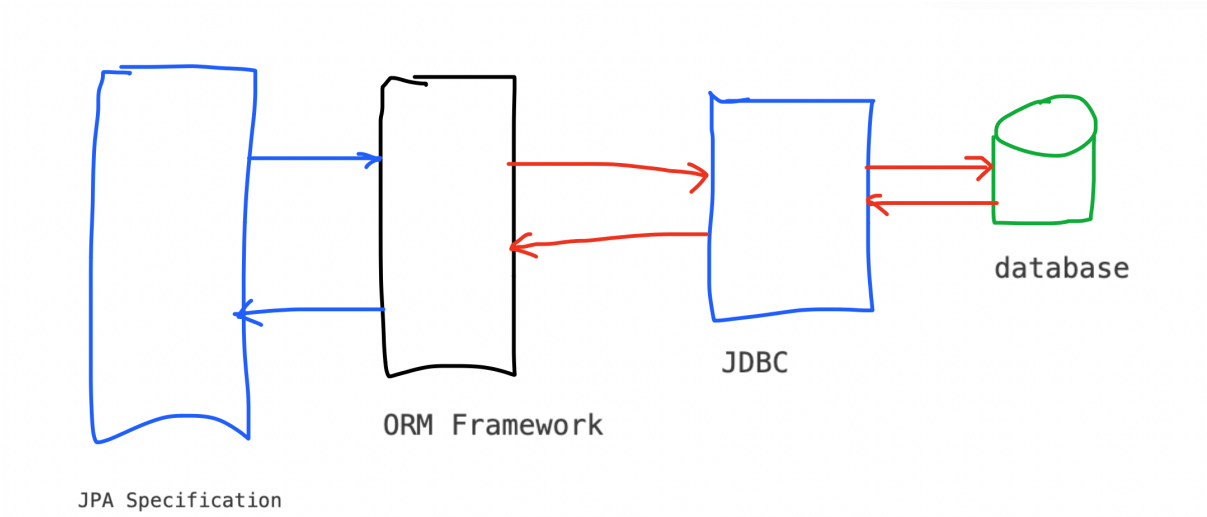
Exp

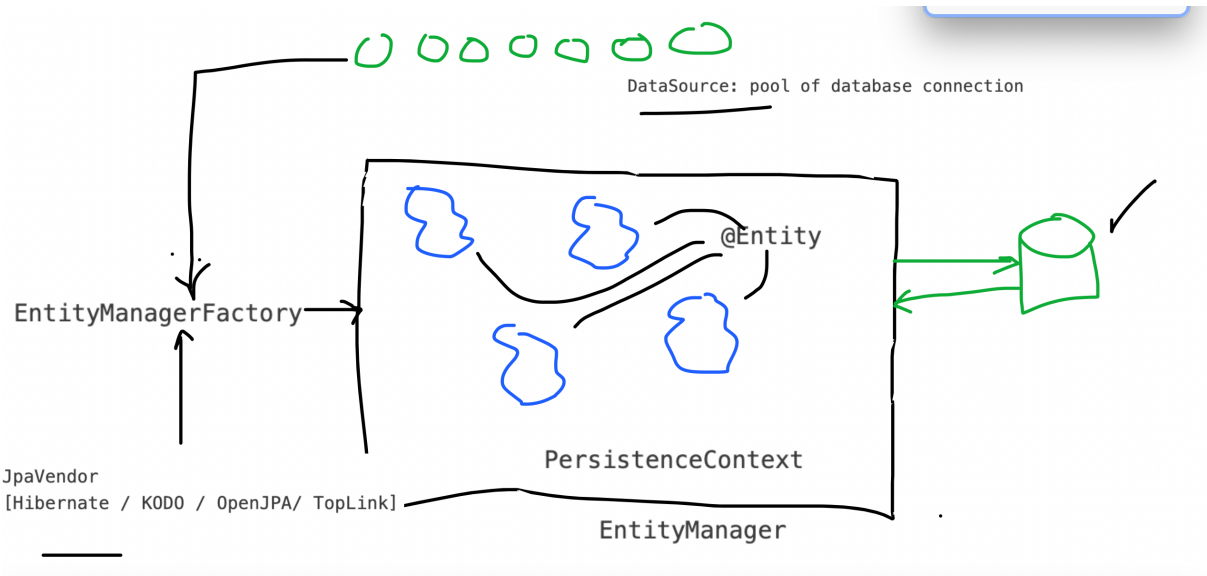
Jdbc

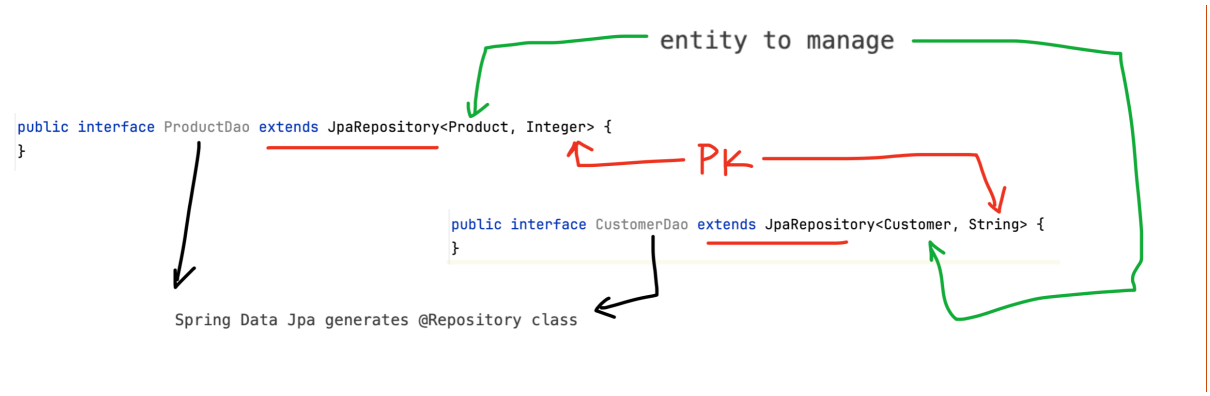
```java
@Override
public void addProduct(Product p) throws PersistenceException{
    Connection con = null;
    try {
        String SQL = "INSERT INTO products (id, name, price) VALUES(0, ?, ?)";
        con = DriverManager.getConnection(URL, USER, PASSWORD);
        PreparedStatement ps = con.prepareStatement(SQL);
        ps.setString(1, p.getName());
        ps.setDouble(2, p.getPrice());
        ps.executeUpdate();
    } catch (SQLException ex) {
        if(ex.getErrorCode() ==1062) {
            throw new PersistenceException("Product with given id: " + p.getId() + " already exists!!", ex);
        } else if(ex.getErrorCode() == 1054) {
            throw new PersistenceException("Unable to add Product!!", ex);
        }

    } finally {
        if(con != null) {
            try {
                con.close();
            } catch (SQLException e) {
                throw new RuntimeException(e);
            }
        }
    }
}
```

ORM

```java
@Override
public void addProduct(Product p) throws PersistenceException{
    em.persist(p);
}
```


database
JDBC
ORM Framework
JPA Specification


DataSource: pool of database connection
@Entity
EntityManagerFactory
PersistenceContext
JpaVendor
[Hibernate / KODO / OpenJPA/ TopLink]
EntityManager


entity to manage
public interface ProductDao extends JpaRepository<Product, Integer> {
}
PK
public interface CustomerDao extends JpaRepository<Customer, String> {
}
Spring Data Jpa generates @Repository class

```
mysql> select * from customers;
+-----------------+-------+-------+
| email           | fname | lname |
+-----------------+-------+-------+
| anna@gmail.com  | Anna  | Ben   |
| roger@gmail.com | Roger | Smith |
+-----------------+-------+-------+
```

```
mysql> select * from products;
+----+----------------+--------+-------+
| id | name           | price  | qty   |
+----+----------------+--------+-------+
|  1 | iPhone 15      |  89000 |  100  |
|  2 | Samsung OLED   | 210000 |  100  |
|  3 | Wacom          |   4500 |  100  |
|  4 | Tata Play      |   5400 |  100  |
|  5 | Logitech Mouse |    540 |  100  |
+----+----------------+--------+-------+
```

orders

oid	order_date	customer_fk	Total
232	20-JUL-2024 4: 30:00	anna@gmail.com	89019.33
233	20-JUL-2024 5:15:21	roger@gmail.com	<<total>>
234	22-JUL-2024 10:30:00	anna@gmail.com	<<total>>

line_items

Item_id	product_fk	order_fk	qty	Amount
898	3	232	2	8000
899	1	232	1	88000.00
900	1	233	2	150000.00